MaskAD: A Hybrid Few-Shot Masked Autoencoder and Siamese Network for Time-Series Anomaly Detection

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I. ABSTRACT

Time-series anomaly detection is an important research area in machine learning with applications in healthcare monitoring, industrial systems, network security, smart city infrastructure, and cyber-physical systems. Traditional supervised learning methods require large amounts of labeled data for training. However, in real-world environments, anomaly labels are rare, expensive, and difficult to collect. This creates a major limitation for conventional supervised anomaly detection models and motivates the need for more efficient data-driven approaches.

This paper presents MaskAD, a hybrid few-shot time-series anomaly detection framework that combines masked autoencoder-based representation learning with Siamese network-based metric learning. The proposed model first learns normal temporal patterns through masked reconstruction. Then, the learned encoder is used in a Siamese network trained with triplet loss to improve separation between normal and anomalous patterns in the embedding space. Finally, a prototype-based distance scoring mechanism is applied to classify anomalies in a robust and scalable manner.

The proposed system is evaluated using the Numenta Anomaly Benchmark dataset. Experimental results show progressive improvement across four model variants. The optimized model achieves a precision of 0.8148, recall of 0.9565, and F1-score of 0.8800. These results demonstrate that the proposed hybrid framework is effective for few-shot anomaly detection in time-series data and suitable for real-world deployment scenarios.

II. KEYWORDS:

Time-series anomaly detection, few-shot learning, masked autoencoder, Siamese network, triplet loss, prototype-based classification, NAB dataset.

III. INTRODUCTION

Time-series anomaly detection refers to the process of identifying abnormal, unusual, or unexpected patterns in sequential data that deviate significantly from normal behavior. It has become a critical task in modern intelligent systems due to the rapid growth of data generated from sensors, devices, and digital platforms. Time-series data is continuously produced in a wide range of real-world domains, including industrial process monitoring, healthcare and patient monitoring systems, cybersecurity and intrusion detection, financial transaction analysis, smart city infrastructure, traffic management systems, and Internet of Things (IoT) applications. In these domains, the timely detection of anomalies is essential, as it can help prevent system failures, reduce operational risks, enhance safety, and support informed decision-making.

Despite its importance, anomaly detection in time-series data presents several significant challenges. One of the primary difficulties is the extreme imbalance between normal and anomalous data. In most real-world scenarios, anomalies are rare events, while the majority of the data represents normal system behavior. This imbalance makes it difficult to apply traditional supervised learning techniques, which typically require a large number of labeled examples from all classes. Moreover, obtaining labeled anomaly data is often expensive, time-consuming, and requires domain expertise, making it impractical in many applications.

Another major challenge arises from the inherent characteristics of time-series data. Unlike static data, time-series observations are sequentially ordered and exhibit complex temporal dependencies. These sequences may contain trends, seasonality, noise, irregular fluctuations, and abrupt changes. As a result, an effective anomaly detection model must be capable of capturing both short-term variations and long-term temporal patterns, while also being robust to noise and data irregularities.

To address these challenges, recent research has focused on leveraging few-shot learning and self-supervised learning techniques. Few-shot learning aims to learn meaningful representations from a very limited number of labeled samples, making it suitable for anomaly detection scenarios where labeled anomalies are scarce. On the other hand, self-supervised learning enables models to learn useful feature representations from unlabeled data by solving auxiliary tasks such as reconstruction, prediction, or masking. These approaches reduce the dependency on labeled data and improve the model's ability to generalize.

Among self-supervised methods, autoencoder-based models have gained significant popularity for anomaly detection. Autoencoders learn to reconstruct normal patterns by compressing input data into a latent representation and then reconstructing it. Anomalies can be detected based on high reconstruction errors. However, these models have limitations, as they may also reconstruct anomalous samples if the anomalies are subtle or share similarities with normal patterns.

Similarly, Siamese networks have been widely used for metric learning, where the goal is to learn similarity relationships between samples. By mapping inputs into an embedding space, Siamese networks can effectively distinguish between similar and dissimilar patterns. However, their performance depends heavily on the availability of well-defined positive and negative sample pairs or triplets. In anomaly detection tasks, especially under few-shot conditions, obtaining meaningful negative samples is challenging.

To overcome these limitations, this paper proposes a hybrid framework that combines the strengths of both reconstruction-based and metric learning approaches. The proposed system, named MaskAD, integrates masked autoencoder learning with Siamese network-based metric learning to achieve robust anomaly detection under limited labeled data conditions. The masked autoencoder is first used to learn rich temporal representations by reconstructing partially masked input sequences, encouraging the model to capture contextual dependencies. The learned encoder is then transferred to a Siamese network trained with triplet loss, which enhances the separation between normal and anomalous samples in the embedding space. Finally, a prototype-based distance scoring mechanism is applied to classify samples based on their deviation from normal behavior.

The main contributions of this paper can be summarized as follows:

- A novel hybrid framework (MaskAD) is proposed that combines masked autoencoder-based representation learning with Siamese network-based metric learning for few-shot time-series anomaly detection.
- A pseudo-negative sampling strategy is introduced to improve the effectiveness of triplet-based metric learning under limited anomaly label conditions.
- A prototype-based anomaly scoring mechanism is developed to enhance interpretability and stability in anomaly detection.
- Multiple model variants are designed and evaluated to demonstrate the progressive improvement and effectiveness of the proposed approach.

Overall, the proposed MaskAD framework addresses key limitations of existing methods and provides a robust, scalable, and efficient solution for anomaly detection in complex time-series data.

IV. Literature Review

Time-series anomaly detection has been extensively studied using a wide range of approaches, including statistical techniques, classical machine learning methods, reconstruction-based models, and modern deep learning architectures. Traditional statistical methods, such as ARIMA and control charts, typically rely on assumptions about the underlying data distribution and stationarity. However, real-world time-series data are often noisy, nonlinear, non-stationary, and dynamically evolving over time, making these assumptions difficult to satisfy. As a result, the effectiveness of purely statistical approaches is often limited in complex real-world scenarios.

To address these challenges, machine learning and deep learning methods have gained significant attention due to their ability to automatically learn meaningful patterns and representations from raw data. In particular, deep learning models such as recurrent neural networks (RNNs), convolutional neural networks (CNNs), autoencoders, and transformer-based architectures have been widely applied to time-series anomaly detection. Among these, autoencoders are especially popular because they learn compact latent representations of normal patterns and identify anomalies based on reconstruction errors. Despite their effectiveness, reconstruction-based approaches sometimes struggle to clearly separate normal and anomalous samples, especially when anomalies are subtle or structurally similar to normal data.

Belton et al. proposed FewSOME, a one-class few-shot anomaly detection method based on Siamese networks. This approach focuses on learning similarity relationships among normal samples by mapping them into an embedding space, where anomalies are detected based on distance metrics. The method demonstrates that Siamese networks can perform effectively even in low-data or few-shot settings. However, its performance heavily depends on the quality and diversity of training pairs or triplets, which can be difficult to construct in practice.

Huang et al. introduced a few-shot anomaly detection framework for HVAC systems using an LSTM-based autoencoder combined with domain adaptation techniques. The LSTM autoencoder effectively captures temporal dependencies and sequential patterns in time-series data, while domain adaptation helps improve generalization across different environments. Although the method shows promising results in specific applications, it still relies on domain-specific adjustments, which may limit its scalability and applicability to diverse datasets.

Zhuang et al. proposed TAMA, a multimodal anomaly detection framework that leverages large-scale multimodal models. In this approach, time-series data are transformed into visual representations, enabling the use of advanced multimodal reasoning techniques. While TAMA achieves strong detection performance, it requires large pretrained models and significant computational resources, making it less suitable for resource-constrained or real-time applications.

Lai et al. developed MOSAD, an open-set multivariate time-series anomaly detection framework that integrates generative, discriminative, and contrastive learning strategies. This hybrid approach allows the system to detect both known and previously unseen anomalies, improving robustness and generalization. However, the increased architectural complexity and training requirements make it more difficult to implement and optimize compared to simpler models.

Fang et al. proposed FastRecon, a few-shot industrial anomaly detection method based on feature reconstruction. Instead of reconstructing raw input data, this method reconstructs feature representations using reference normal samples. Anomalies are identified based on reconstruction discrepancies in the feature space. While FastRecon is computationally efficient and effective in industrial settings, its performance largely depends on the quality of pretrained feature extractors.

From the reviewed literature, several key limitations can be identified. First, many existing approaches rely heavily on pretrained feature extractors or large-scale models, which increases computational cost and reduces flexibility. Second, methods based solely on reconstruction error may fail to provide clear separation between normal and anomalous data, particularly in complex scenarios. Third, metric-learning approaches, such as Siamese or triplet networks, require carefully designed negative samples, which are not always readily available. Finally, many advanced models are computationally expensive and may not be suitable for real-time or resource-constrained environments.

These challenges highlight the need for a more efficient and robust approach. Therefore, this study proposes a hybrid framework that integrates reconstruction learning, metric learning, and prototype-based classification to achieve improved anomaly detection performance while maintaining computational efficiency.

V. Methodology

The proposed MaskAD framework is designed to address the challenges of few-shot time-series anomaly detection by combining self-supervised representation learning with metric learning and prototype-based scoring. The overall system consists of multiple stages, including data preprocessing, sliding window generation, masked autoencoder training, Siamese metric learning, prototype construction, anomaly scoring, thresholding, and final evaluation. Each component plays a critical role in improving the robustness and effectiveness of anomaly detection under limited labeled data conditions.

System Overview

The overall anomaly detection pipeline is illustrated in Fig. 1. Initially, raw time-series data are preprocessed to remove noise, handle missing values, and normalize the input. The cleaned data are then segmented into overlapping windows using a sliding window approach, which enables the model to process temporal sequences in fixed-size inputs.

These windowed samples are first fed into a masked autoencoder, which learns compact and meaningful latent representations of normal temporal behavior through a self-supervised reconstruction task. The trained encoder is subsequently transferred to a Siamese network, where metric learning is applied using triplet loss to enhance the separation between normal and anomalous samples in the embedding space. Finally, a prototype-based scoring mechanism is used to compute anomaly scores by measuring the distance between sample embeddings and a learned normal prototype.

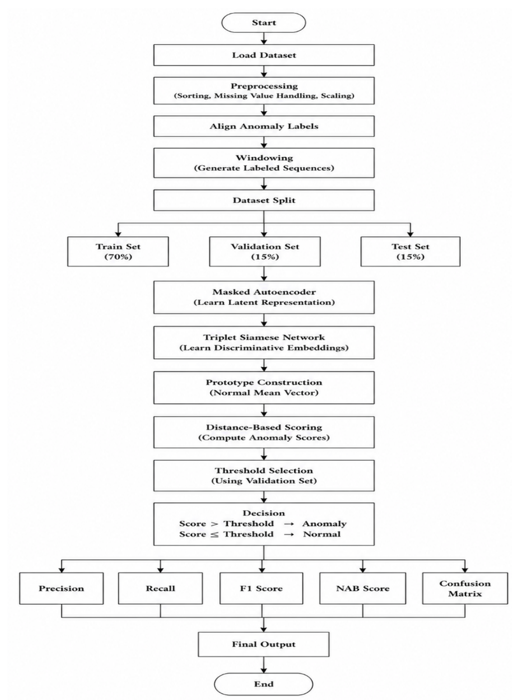

Fig. 1. End-to-end anomaly detection pipeline.

The detailed architecture of the proposed MaskAD framework is shown in Fig. 2. By integrating reconstruction-based learning with similarity-based learning, the framework effectively captures both temporal patterns and relational structures in the data, leading to improved anomaly detection performance in few-shot scenarios.

Data Preprocessing

The dataset is first sorted in chronological order based on timestamps to preserve temporal consistency. Missing values, which are common in real-world time-series data, are handled using a combination of interpolation, forward filling, and backward filling techniques. These methods ensure continuity in the data while minimizing distortion of underlying patterns.

After handling missing values, standard normalization is applied to scale the data to zero mean and unit variance. The normalized value x' is calculated as:

$$x' = \frac{x - \mu}{\sigma} \tag{1}$$

where x is the original value, μ is the mean, and σ is the standard deviation. Normalization improves numerical stabil-

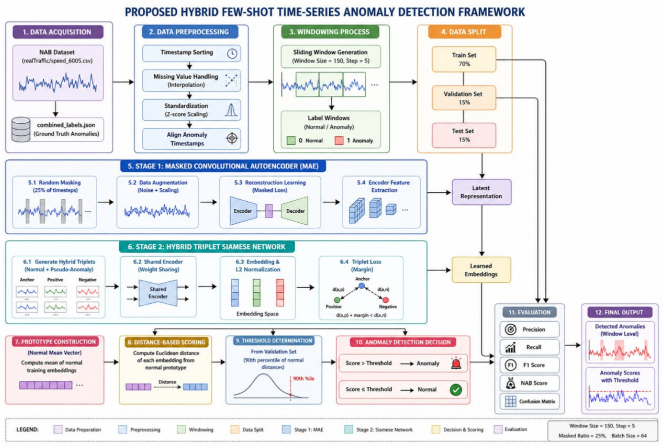

Fig. 2. Overview of the proposed MaskAD framework.

ity, accelerates convergence during training, and ensures that features contribute equally to the learning process.

Sliding Window Generation

To convert the continuous time-series into a format suitable for neural networks, a sliding window technique is applied. This method segments the sequence into overlapping windows of fixed length. Each window is defined as:

$$W_t = \{x_t, x_{t+1}, x_{t+2}, \ldots, x_{t+L-1}\} \tag{2}$$

where L denotes the window size.

The use of overlapping windows increases the number of training samples and allows the model to capture local temporal patterns effectively. Each window is assigned a binary label: it is labeled as anomalous if it contains at least one anomaly point, and normal otherwise. This transformation enables batch-based training and facilitates supervised evaluation.

Masked Autoencoder

The masked autoencoder is employed to learn robust representations of normal time-series behavior through a self-supervised learning objective. During training, a random subset of the input window is masked, and the model is tasked with reconstructing the original unmasked signal. This masking strategy prevents the model from simply memorizing the input and encourages it to learn meaningful temporal dependencies and contextual relationships.

The reconstruction loss is computed using mean squared error:

$$\mathcal{L}_{rec} = \frac{1}{N}\sum_{i=1}^{N} \|x_i - \hat{x}_i\|^2 \tag{3}$$

where x_i is the original input, $\hat{x}_i$ is the reconstructed output, and N is the number of samples. The encoder part of the trained autoencoder is then used as a feature extractor, providing latent embeddings for the subsequent metric learning stage.

Siamese Network with Triplet Loss

To enhance the discriminative power of the learned embeddings, a Siamese network is applied using triplet loss. The network takes three inputs: an anchor sample (a), a positive sample (p) that is similar to the anchor, and a negative sample (n) that is dissimilar.

The triplet loss function is defined as:

$$\mathcal{L}_{triplet} = \max(0, d(a,p) - d(a,n) + m) \tag{4}$$

where $d(\cdot)$ represents the Euclidean distance between embeddings, and m is a margin parameter that enforces a minimum separation between positive and negative pairs.

This loss function encourages the model to minimize the distance between similar samples while maximizing the distance between dissimilar samples. As a result, normal samples form compact clusters in the embedding space, while anomalous samples are pushed further away, improving separability and detection performance.

Prototype-Based Anomaly Scoring

After training the Siamese network, embeddings of normal samples are used to compute a prototype vector representing the center of normal behavior:

$$c = \frac{1}{N}\sum_{i=1}^{N} z_i \tag{5}$$

where z_i denotes the embedding of a normal sample.

For a given test sample, the anomaly score is calculated as the Euclidean distance between its embedding and the prototype:

$$S(x) = \|z(x) - c\|_2 \tag{6}$$

where $z(x)$ is the embedding of the test sample. A higher score indicates a greater deviation from normal behavior.

Thresholding and Decision Making

To classify samples as normal or anomalous, a threshold is applied to the anomaly scores. If $S(x)$ exceeds the threshold, the sample is classified as anomalous; otherwise, it is considered normal. In this work, a percentile-based threshold is used for simplicity and consistency across experiments.

This thresholding mechanism converts continuous anomaly scores into binary decisions, enabling straightforward evaluation using standard metrics. Although effective, the choice of threshold plays a critical role in balancing detection sensitivity and false alarm rate, and can be further optimized in future work.

Overall, the proposed methodology integrates multiple complementary techniques to address the limitations of existing approaches. By combining masked reconstruction, metric learning, and prototype-based scoring, the MaskAD framework achieves robust and efficient anomaly detection in few-shot time-series settings.

VI. Results

Experimental Setup

All experiments are implemented using Python with TensorFlow/Keras as the primary deep learning framework. NumPy and Pandas are utilized for efficient data preprocessing and manipulation, while Scikit-learn is used for normalization, dataset splitting, and performance evaluation. Matplotlib is employed to visualize anomaly scores and model behavior across different experiments. The Adam optimizer is used for training due to its adaptive learning rate and stable convergence properties.

The experiments are conducted under a consistent set of configurations to ensure fair comparison among models. The common experimental settings are summarized as follows:

- Dataset: Numenta Anomaly Benchmark (NAB)
- Optimizer: Adam
- Loss functions: masked mean squared error for reconstruction and triplet loss for metric learning
- Evaluation metrics: precision, recall, and F1-score
- Input format: sliding window-based time-series samples
- Training strategy: few-shot learning with pseudo-negative sampling

Dataset Description

The Numenta Anomaly Benchmark (NAB) dataset consists of real-world time-series data collected from various domains, including server metrics, system performance logs, and sensor readings. It is widely used for evaluating anomaly detection algorithms due to its realistic characteristics, such as streaming data patterns, temporal dependencies, and sparsely occurring anomaly events.

In this work, the raw time-series data are first normalized to ensure stable training and prevent scale-related bias. The anomaly labels provided in the dataset are aligned with their corresponding timestamps. A sliding window technique is then applied to convert the sequential data into fixed-length samples suitable for deep learning models. Each window is assigned a binary label based on whether it contains at least one anomalous point, enabling supervised evaluation of detection performance.

Evaluation Metrics

The performance of the proposed models is evaluated using precision, recall, and F1-score, which are standard metrics for anomaly detection tasks.

Precision is defined as:

$$Precision = \frac{TP}{TP + FP} \tag{7}$$

Recall is defined as:

$$Recall = \frac{TP}{TP + FN} \tag{8}$$

F1-score is defined as:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{9}$$

where TP (true positives) represents correctly detected anomalies, FP (false positives) represents incorrectly detected anomalies, and FN (false negatives) represents missed anomaly events. These metrics provide a balanced evaluation by considering both detection accuracy and error rates.

Model 1: Baseline Hybrid Autoencoder with Siamese Network

Model 1 serves as the baseline hybrid architecture, combining a masked convolutional autoencoder with a Siamese network for metric learning. The autoencoder learns to reconstruct normal patterns, while the Siamese network enforces similarity constraints in the latent space. Pseudo-negative samples are generated by selecting the top 5% of windows with the highest reconstruction errors.

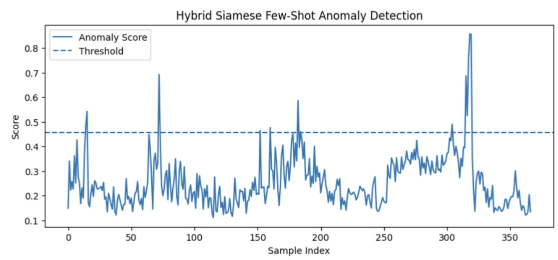

Fig. 3. Anomaly scores for Model 1.

As illustrated in Fig. 3, only a limited number of anomaly score peaks exceed the detection threshold. This indicates weak separation between normal and anomalous samples. The poor performance can be attributed to insufficient diversity in pseudo-negative samples, which limits the effectiveness of triplet-based metric learning.

Model 2: Improved Pseudo-Negative Sampling

Model 2 enhances the baseline by introducing a more diverse pseudo-negative sampling strategy. In addition to high reconstruction error windows, randomly selected samples are included to form more varied triplets during training.

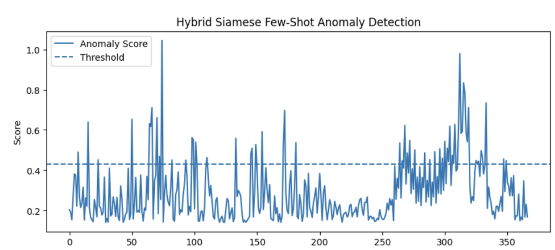

Fig. 4. Anomaly scores for Model 2.

As shown in Fig. 4, the anomaly score distribution becomes more distinguishable, with a greater number of peaks crossing the threshold. This improvement indicates that increased diversity in negative samples helps the model learn a more discriminative embedding space. However, some overlap between normal and anomalous regions still persists.

Model 3: Residual Autoencoder with Augmented Siamese Learning

Model 3 introduces a deeper residual convolutional autoencoder, which improves representation learning by facilitating better gradient flow and reducing vanishing gradient issues. Additionally, data augmentation techniques such as Gaussian noise injection and scaling are applied to improve robustness. The window size is increased to capture longer temporal dependencies.

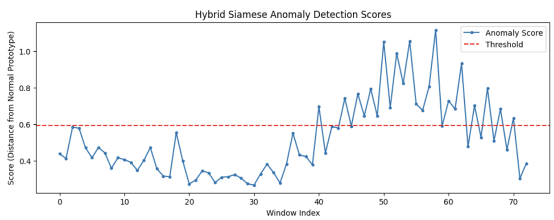

Fig. 5. Anomaly scores for Model 3.

Fig. 5 demonstrates that anomaly regions become more clearly distinguishable compared to previous models. The combination of residual learning and data augmentation enhances the model's ability to generalize and capture complex temporal patterns, leading to a significant performance improvement.

Model 4: Optimized Hybrid System

Model 4 represents the final optimized framework, integrating all improvements from previous models. It employs a larger window size, a stronger residual encoder, hybrid triplet sampling, and prototype-based anomaly scoring. The prototype-based approach further refines anomaly detection by comparing samples against learned normal prototypes.

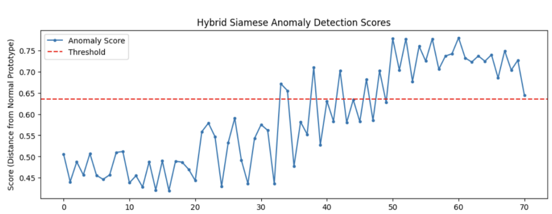

Fig. 6. Anomaly scores for Model 4.

As shown in Fig. 6, the anomaly scores exhibit clear separation between normal and anomalous regions. Most anomaly segments exceed the threshold, while normal regions remain consistently below it. This indicates that the model achieves strong discrimination and stability.

Quantitative Results

The quantitative comparison of all models is presented in Table I.

TABLE I
PERFORMANCE COMPARISON OF PROPOSED MODELS

Model	Precision	Recall	F1-Score
Model 1	0.0770	0.0170	0.0270
Model 2	0.3390	0.3330	0.3360
Model 3	0.6818	0.7500	0.7143
Model 4	0.8148	0.9565	0.8800

The results demonstrate a consistent improvement across successive models. Model 1 shows very low performance due to weak anomaly separation. Model 2 achieves moderate improvement by enhancing negative sampling. Model 3 provides a substantial performance boost through residual learning and augmentation. Finally, Model 4 achieves the best performance, with high precision, recall, and F1-score, indicating strong detection capability and robustness.

VII. Discussion

The experimental findings clearly demonstrate the effectiveness of the proposed hybrid framework for few-shot time-series anomaly detection. The systematic progression from Model 1 to Model 4 highlights the importance of integrating multiple learning paradigms, including reconstruction learning, metric learning, and prototype-based scoring. Each successive model introduces targeted improvements that collectively enhance anomaly discrimination and overall system robustness.

Model 1 serves as a baseline and reveals key limitations of using simple pseudo-negative sampling based solely on reconstruction error. The lack of diversity in negative samples leads to poor embedding separation, making it difficult for the Siamese network to learn meaningful distance relationships. This result emphasizes that reconstruction error alone is not sufficient to guide effective metric learning, particularly in few-shot scenarios where labeled anomalies are scarce.

Model 2 addresses this limitation by incorporating randomly selected samples into the pseudo-negative set, thereby increasing diversity. This modification leads to noticeable performance improvement, confirming that the quality and variability of training samples are critical factors in metric learning. However, the remaining overlap between normal and anomalous samples indicates that sampling strategies alone are not enough to achieve optimal performance.

Model 3 introduces a deeper residual convolutional autoencoder along with data augmentation techniques such as Gaussian noise injection and scaling. The residual architecture improves gradient flow and enables the model to learn richer hierarchical representations, while augmentation enhances generalization by exposing the model to variations of normal patterns. Additionally, increasing the window size allows the model to capture longer temporal dependencies, which is essential for identifying anomalies that span extended time intervals. These combined improvements result in a substantial gain in detection performance.

Model 4 represents the final optimized system and achieves the best results by integrating all previous enhancements with a prototype-based scoring mechanism. This approach introduces an additional layer of interpretability and stability by comparing latent representations against learned prototypes of normal behavior. As a result, anomaly detection becomes more robust to noise and variation in the data. The high recall achieved by Model 4 indicates that the framework is highly effective in identifying anomalous events, minimizing the risk of missed detections. This is particularly important in safety-critical applications such as healthcare monitoring, industrial fault detection, financial fraud detection, and cybersecurity systems.

Despite the strong performance, it is important to consider the trade-off between recall and precision. While high recall ensures that most anomalies are detected, lower precision may lead to an increased number of false alarms. In real-world deployments, excessive false positives can reduce user trust and increase operational costs. Therefore, achieving a balanced trade-off between precision and recall is essential for practical usability. Techniques such as cost-sensitive learning or post-processing filters could be explored to further refine this balance.

Another key factor influencing performance is the selection of the anomaly detection threshold. In this study, a percentile-based thresholding approach is adopted due to its simplicity and ease of implementation. However, fixed thresholds may not be optimal for non-stationary or dynamically changing time-series data. Adaptive thresholding methods, such as those based on statistical process control or online learning, could provide more flexible and context-aware anomaly detection.

From a computational perspective, the proposed hybrid framework maintains a reasonable balance between performance and efficiency. While deeper architectures and additional components increase model complexity, the use of sliding windows and efficient training strategies ensures scalability to real-world datasets. Nevertheless, further optimization may be required for deployment in resource-constrained environments or real-time applications.

Overall, the results confirm that combining masked reconstruction learning with Siamese-based metric learning and prototype-based scoring significantly enhances both representation quality and anomaly discrimination. The hybrid design effectively addresses the limitations of individual approaches and provides a more robust solution for few-shot anomaly detection. These findings suggest that integrating complementary learning strategies is a promising direction for future research in time-series anomaly detection.

VIII. Conclusion

This paper presents MaskAD, a novel hybrid framework for few-shot time-series anomaly detection that integrates masked autoencoder-based representation learning with Siamese network-based metric learning. The masked autoencoder is designed to learn meaningful and compact representations of normal temporal patterns through partial input reconstruction, enabling the model to capture underlying data structures even in the presence of noise. At the same time, the Siamese network introduces a metric learning component that enhances the separation between normal and anomalous samples in the latent embedding space, leading to improved anomaly discrimination.

The proposed framework is evaluated on the Numenta Anomaly Benchmark (NAB) dataset, which provides realistic and diverse time-series data with labeled anomaly events. To systematically analyze the effectiveness of the approach, four model variants are developed, each incorporating incremental improvements in architecture design, sampling strategy, and training methodology. The experimental results demonstrate a clear and consistent performance improvement across the models, highlighting the contribution of each component in the overall framework.

In particular, the final optimized model (Model 4) achieves the best performance, with a precision of 0.8148, recall of 0.9565, and F1-score of 0.8800. These results indicate that the proposed approach is highly effective in detecting anomalies while maintaining a good balance between detection accuracy and false alarm rate. The high recall value further confirms the model's capability to identify the majority of anomalous events, which is critical in applications where missed detections can have serious consequences.

The findings of this study demonstrate that combining reconstruction-based learning with metric learning provides a powerful solution for anomaly detection, especially in few-shot scenarios where labeled anomaly data is scarce or unavailable. Unlike traditional methods that rely solely on reconstruction error or predefined statistical assumptions, the proposed hybrid framework leverages both feature learning and similarity-based reasoning to achieve robust performance.

In addition to its strong detection capability, the proposed system maintains reasonable computational efficiency, making it suitable for practical deployment in real-world environments. Potential application domains include industrial process monitoring, healthcare and patient monitoring systems, smart city infrastructure, financial fraud detection, and cybersecurity threat analysis, where early and accurate anomaly detection is essential.

Despite the promising results, there are several directions for future work. First, adaptive thresholding techniques could be explored to improve performance in dynamic and non-stationary environments. Second, incorporating attention mechanisms or transformer-based encoders may further enhance the model's ability to capture long-range temporal dependencies. Third, extending the framework to fully unsupervised or online learning settings could improve scalability and real-time applicability.

Overall, this work demonstrates that the integration of masked reconstruction learning, Siamese metric learning, and prototype-based reasoning offers a robust and effective solution for few-shot time-series anomaly detection, and provides a strong foundation for future research in this area.

IX. Future Work

Although the proposed MaskAD framework demonstrates strong performance for few-shot time-series anomaly detection, several directions can be explored to further enhance its effectiveness, scalability, and applicability in real-world scenarios.

First, transformer-based architectures can be investigated to better capture long-range temporal dependencies in time-series data. Unlike convolutional or recurrent models, transformers rely on self-attention mechanisms that allow the model to learn global relationships across time steps. Integrating transformer encoders into the current framework may significantly improve the representation of complex temporal patterns, especially in datasets with long-term dependencies and irregular structures.

Second, attention mechanisms can be incorporated into the existing architecture to enable the model to focus on the most informative time steps within each input window. By assigning higher importance to critical segments of the time-series, attention-based models can improve both reconstruction quality and anomaly discrimination. This could be particularly useful in scenarios where anomalies are subtle or localized within specific regions of the sequence.

Third, semi-supervised learning strategies can be explored to leverage a small number of labeled anomaly samples. In practical applications, obtaining fully labeled datasets is often expensive and time-consuming. By incorporating a limited amount of labeled anomalies during training, the model can improve the quality of pseudo-negative samples and learn more discriminative decision boundaries. Techniques such as contrastive learning with labeled anchors or hybrid supervised-unsupervised objectives could further enhance performance.

Another promising direction is the development of adaptive thresholding techniques for anomaly detection. In this study, a fixed percentile-based threshold is used for simplicity; however, real-world time-series data are often non-stationary and subject to distributional changes over time. Dynamic thresholding methods that adjust based on recent data statistics, seasonal trends, or online learning mechanisms could significantly improve robustness and reduce false positives in evolving environments.

In addition, model efficiency and deployment considerations should be addressed. While the proposed framework achieves strong performance, its computational complexity may limit its use in resource-constrained environments. Techniques such as model pruning, weight quantization, and knowledge distillation can be applied to reduce model size and inference latency without significantly sacrificing accuracy. These optimizations are particularly important for deploying anomaly detection systems on edge devices or in real-time monitoring applications.

Furthermore, future work could explore extending the framework to fully unsupervised or online learning settings. An online learning approach would allow the model to continuously update its knowledge based on incoming data streams, making it more adaptable to concept drift and evolving anomaly patterns. This would be highly beneficial for applications such as network monitoring, IoT systems, and industrial automation.

Finally, evaluating the proposed framework on a wider range of benchmark datasets and real-world applications would provide deeper insights into its generalization capability. Cross-domain validation, including domains such as finance, healthcare, and environmental monitoring, could further demonstrate the robustness and versatility of the approach.

Overall, these future research directions aim to enhance the flexibility, scalability, and real-world applicability of the proposed MaskAD framework, paving the way for more advanced and practical time-series anomaly detection systems.

References

[1] N. Belton, A. F. Smeaton, and S. Little, "FewSOME: One-class few-shot anomaly detection using Siamese networks," in *Proc. International Conference on Pattern Recognition Applications and Methods*, 2023.

[2] Y. Huang, J. Wang, and X. Li, "Few-shot anomaly detection for HVAC systems using LSTM autoencoder with domain adaptation," *Energy and Buildings*, 2022.

[3] Y. Zhuang, J. Zhang, and L. Wang, "TAMA: Multimodal time series anomaly detection using large multimodal models," 2024.

[4] K. Lai, Y. Chen, and M. Zhang, "MOSAD: Open-set multivariate time-series anomaly detection," 2023.

[5] Z. Fang, X. Liu, and J. Han, "FastRecon: Few-shot industrial anomaly detection via feature reconstruction," 2023.

[6] A. Lavin and S. Ahmad, "Evaluating real-time anomaly detection algorithms: The Numenta Anomaly Benchmark," in *Proc. IEEE International Conference on Machine Learning and Applications*, 2015.

[7] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. International Conference on Learning Representations*, 2015.

[8] G. E. Hinton and R. R. Salakhutdinov, "Reducing the dimensionality of data with neural networks," *Science*, vol. 313, no. 5786, pp. 504–507, 2006.

[9] J. Bromley, I. Guyon, Y. LeCun, E. Sackinger, and R. Shah, "Signature verification using a Siamese time delay neural network," in *Proc. Advances in Neural Information Processing Systems*, 1993.

[10] F. Schroff, D. Kalenichenko, and J. Philbin, "FaceNet: A unified embedding for face recognition and clustering," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition*, 2015.